



Polytension Locomotion

TEE Multi-System Locomotives

This report aims to showcase the most important multi-system locomotives of the TEE era in model form. Multi-system vehicles are locomotives and multiple units that can operate on more than one electrical system. In Europe, four main electrical systems are in use and in operation:

- DC 1500 V (Netherlands, France)
- DC 3000 V (Belgium, Italy)
- AC 15 kV, 16 2/3 Hz (Switzerland, Germany, Austria)
- AC 25 kV, 50 Hz (France, Luxembourg)

With the expansion of the electrified TEE network in Europe to the borders of France, Belgium, and the Netherlands, the need arose for a locomotive that could operate universally on all four electrical systems. One of the first was the so-called European locomotive, the Class 184.

The E410 (184) "Europa locomotive," built by Krupp, of which five were delivered in two variants (AEG: 184 001–003 and BBC: 184 111–112), was demanding in terms of maintenance and operation even in its day.

The 184 was notable for its high susceptibility to malfunctions, as its operation on the DC lines in Belgium (3 kV nominal, with voltage peaks up to 6 kV!) could not be reliably controlled from a technical standpoint.



Polytension Loks in H0: Märklin SBB RAe 1051 – Lemaco SNCF CC 40110 – L.S.Models DB BR 184 111-3 – Märklin SNCB BB 1606

The special three-phase AC technology, pioneering at the time, with water-cooled converters and a current-guided intermediate circuit of the Europalok locomotive, was based by AEG on their multi-system locomotive 182 001. Along with the distinctive four-system locomotives E 410 / BR 184, four new two-system locomotives were also built as E310 (181). Of particular interest is the Märklin Insider model 181 203 from 2023, which operated as the TEE Goethe or Mozart:



Piko Br 181 Luxembourg – Märklin Br 181.2 Goethe Insider

The special three-phase AC technology, pioneering work at the time, with water-cooled converters and current-guided wiring, was truly exciting, in the truest sense of the word. The system change was also thrilling: at the border, the locomotive driver lowered the German pantograph, overcame a section without current with controlled momentum, and finally switched on the French pantograph. The elegant, flat locomotive body was a consequence of the lower-lying French overhead lines. The fascinating 181.2 and the BR 184 have made European TEE history.

During the TEE era, exactly three railway companies were involved in the development of Polytension locomotives: DB, SNCB, and SNCF¹. Diesel locomotives were also in use from 1957 onwards, most notably the TEE Talgo, which could also handle gauge changes. The articulated trains passed through an automatic gauge-changing facility at the border! The TEE Catalan Talgo was an international passenger train that connected Geneva and Barcelona from 1969 onwards.

The Swiss Federal Railways (SBB), as the fourth nation (without a multi-system locomotive), developed its legendary six-car electric multiple unit, the RAe TEE II, which was designed for four different railway electrification systems and

¹ In a narrower sense, diesel locomotives can also be considered multi-system locomotives.

entered service in 1961 as an electrotechnical pioneer. At that time, the SBB did not have a true multi-system locomotive in operation. It was also the first train in Europe that could operate freely under all four European electrical systems. One of the famous TEE Gottardo's power cars, RAe TEE II 1053, still exists today in operational condition under the management of SBB Historic.



TEE Gottardo Einfahrt Zürich HB - 2021

Let's take a look at Belgium and France. The Belgian national railway SNCB introduced its Type 150 locomotives (now designated as Class 15) in 1962, designed for three different voltages. These were followed in 1966 by the four-voltage Type 160 (Class 16) and in 1973 by the Class 18 (180 km/h), of which six were built. The French national railway SNCF also developed and introduced ten four-voltage locomotives of the CC 40100 series between 1964 and 1970.



TEE CC 40105-40109-40108 Märklin, Trix – Polytension Locs

The CC 40100 series consists of ten quadricurrent electric locomotives built by SNCF. They were delivered in two phases: the first four in 1964 and the last six between 1969 and 1970. At 22.03 meters long, they are considered the longest TEE locomotives (along with the CC 18 at **22.08** meters!). The powerful dual-system CC 21000 locomotives (built from the CC 6500 series) are also the heaviest TEE locomotives, weighing in at **122** tons and measuring 20.19 meters.

From 1995 onwards, they were converted to the DC variant CC6500 and assigned to this series (running numbers 6575 to 6578). In 2007, the last CC6500 locomotives were retired, including locomotive 6575, which was preserved in working order, is based in Nîmes, and once again bears the number 21001. The CC 40101 (Alsthom, 1964) can be admired at the Cité du Train in Mulhouse and, along with the SBB TEE Gottardo and the DB Europalok BR 184, is considered another pioneer of a multi-system locomotive for four electrical systems (quadricourant).



https://teetrain2.wordpress.com/wp-content/uploads/2025/12/ctd_cc40010_20250318_133025.jpg?w=768

Conclusion

Multi-system technology played a crucial role in the TEE concept, but in its early days, it was implemented primarily through diesel multiple units (Parsifal, Helvetia, Mediolanum) rather than locomotives.

The RAe TEE II trains are particularly well-known as pioneers: electric multi-system multiple units capable of operating on four different electrical systems across almost the entire TEE network (with the exception of Italian three-phase AC lines). With the further development of multi-system locomotives, operators such as SNCF (CC 40100), SNCB (CC 18), and DB (BR 184) deployed such locomotives to enable through TEE services between France, the Benelux countries, and Germany.

Max Kleiner, Text & Pictures, 16. Februar 2026

Ref: [Polytension – Trans Europ Express](#)

[TEE-Time – Breitschblog](#)

Image credit:

1. 4 Polytension Loks Model

https://teetrain2.wordpress.com/wp-content/uploads/2025/11/tee_polytension_20251124_111623.jpg?w=2048

2. 2 Insider Loks Model

https://teetrain2.wordpress.com/wp-content/uploads/2026/02/br181_insider_20260213_134033.jpg?w=2048

3. TEE Gottardo Image

https://my6.code.blog/wp-content/uploads/2021/06/20210626_tee_zurich_entry.jpg?w=1440

4. 3 CC 40100 Loks Model

https://teetrain2.wordpress.com/wp-content/uploads/2026/01/tee_cc40100_20260116_124652.jpg?w=2048

5. 1 CC 40101 Image

https://teetrain2.wordpress.com/wp-content/uploads/2025/12/ctd_cc40010_20250318_133025.jpg?w=768

Appendix: **TEE Modellbahn Datenbank**

[Trans Europ Express TEE Models](#)

Appendix 2:

TEE Gottardo Cockpit



TEE Gottardo RAe II

[max_tee_cockpit.jpg \(1440×960\)](#)

AI (stable diffusion) generated TEE multi-system locomotive:

